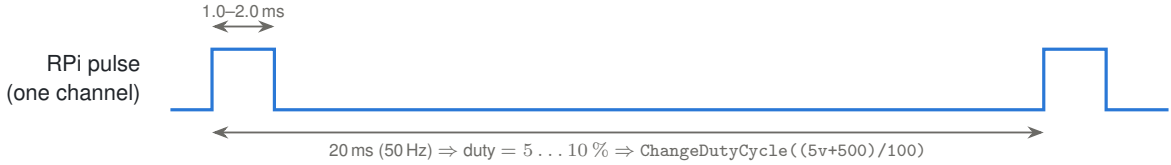
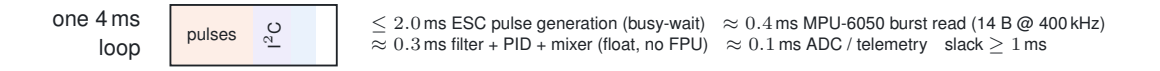

1.0–2.0 ms
RPi pulse
(one channel)
20 ms (50 Hz) ⇒ duty = 5 . . . 10 % ⇒ ChangeDutyCycle((5v+500)/100)

1.1–2.0 ms
ESC pulse
(Uno D4)
4 ms


one 4 ms
loop
pulses
I²C
≤ 2.0 ms ESC pulse generation (busy-wait)
≈ 0.4 ms MPU-6050 burst read (14 B @ 400 kHz)
≈ 0.3 ms filter + PID + mixer (float, no FPU)
≈ 0.1 ms ADC / telemetry
slack ≥ 1 ms